

Quantum Field Theory I Exercise 10

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1 The Beta Function of Scalar Electrodynamics

We consider the theory

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 - \frac{1}{2}(\partial_\mu A^\mu)^2 + (D^\mu \phi)^* (D_\mu \phi) - m^2 \phi^* \phi - \frac{\lambda}{4}(|\phi|^2)^2 \quad (1)$$